

The Thermodynamics of Algorithmic Forgetting

Unifying Physical Emanations, Precision Loss, and Reversible Spacetime

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Introduction: The Isomorphism of Dissipation and Information Erasure

The classical separation between physical thermodynamics, cryptographic side-channel analysis, and numerical linear algebra relies on a set of fundamental, yet ultimately porous, abstractions. Physical heat is traditionally viewed by classical mechanics as the random, unstructured kinetic energy of particles interacting in a system. Electromagnetic emanations are treated as mere hardware artifacts—unintended consequences of electrical currents navigating silicon pathways. Furthermore, numerical precision loss is categorized strictly as an unavoidable mathematical error governed by the finite constraints of floating-point architectures. However, a rigorous analysis of irreversible computation, grounded in Landauer's principle and the observed properties of the Stieltjes moment pipeline, reveals that these seemingly disparate phenomena are exactly isomorphic. Heat, particularly in the context of information processing, is not an unstructured void of thermal ash. Rather, it is the highly structured exhaust of information erasure—the physical and mathematical act of a system actively forgetting its prior geometric constraints.

When a computation forces a physical or algorithmic state to collapse its distinctions—such as when a matrix factorization drops below its working precision floor and truncates a floating-point sequence—the resulting "garbage" is not random. It is a deterministic byproduct of the object's initial conditions. This insight profoundly bridges the physical and the informational realms. Just as electromagnetic fluctuations can broadcast human keystrokes via Van Eck phreaking, or acoustic emissions from a CPU voltage regulator can leak highly secure cryptographic keys to an external observer, the sub-floor algorithmic noise of a standard matrix factorization broadcasts the precise spatial geometry of its underlying support nodes.

This report presents an exhaustive synthesis of the mechanisms by which information loss manufactures the perceived arrow of time, strictly bounds computational energy efficiency, and creates deeply legible side channels that defy traditional models of secure computation. By examining the recent Christoffel Rank Read framework, the thermodynamic bounds of computation, and the deterministic mathematical properties of the LDL^T matrix factorization, it is possible to demonstrate that the universe's native, unobserved state is a

single, reversible, co-present block of information. The passage of time—defined as the sequential progression of distinct algorithmic states—is an instrumental illusion generated entirely by the thermodynamic "tax" paid at the precision floor. Heat, therefore, is not merely a byproduct of computation; it is the fundamental exhalation that makes the next computational cycle physically and mathematically possible.

The Thermodynamic Imperative: Landauer's Limit and Reversible States

To understand the exact nature of algorithmic noise, one must first establish the ultimate physical limits of computation. Landauer's principle, introduced by physicist Rolf Landauer in 1961, dictates that any logically irreversible manipulation of information—such as the erasure of a memory bit or the merging of two distinct computational paths into one—must be accompanied by a corresponding and inevitable increase in the entropy of the surrounding environment. Specifically, the minimum energy required to erase one bit of information is mathematically defined as $kT\ln(2)$, where k represents the Boltzmann constant and T signifies the absolute temperature of the system. At standard room temperature, this thermodynamic cost constitutes an energy dissipation of approximately 2.9×10^{-2} Joules, or 0.018 electron volts.

A device, system, or algorithm is deemed logically irreversible if its original input state cannot be uniquely and perfectly deduced from its final output state. In the context of a physical state matrix, this logical irreversibility occurs precisely when the determinant collapses to zero ($\det = 0$). Experimental setups, such as those utilizing a single colloidal particle trapped within a modulated bistable double-well potential, have successfully demonstrated this limit in a laboratory setting. By lowering the central barrier and applying a tilting force, researchers established that the mean dissipated heat saturates exactly at the Landauer bound in the limit of long, quasi-static erasure cycles. This confirms that the thermodynamic limit is an unavoidable physical reality, independent of the specific device, circuit, or material used to implement the operation.

Conversely, when a system maintains full logical reversibility ($\det \neq 0$), the transformation remains strictly bijective from end to end. According to Charles Bennett's foundational principles of reversible computing, such a system can theoretically execute arbitrary logical operations but strictly cannot discard any intermediate history. Because there is no determinant collapse and no erasure of state, the system never pays the Landauer cost and dissipates exactly zero heat. However, the structural cost of this reversibility is absolute memory retention; the computation must unspool without ever losing its algebraic rank.

Modern commercial processors operate at energy scales roughly a billion times higher than the theoretical Landauer limit, dissipating approximately 5×10^{-13} Joules per 16-bit floating-point operation. However, as computational frameworks relentlessly scale and biological or neuromorphic hardware designs attempt to approach true thermodynamic parity, understanding the structure of sub-Landauer patterns becomes critically important. When a digital system resolves an arbitrary-precision mathematical object into a fixed-precision binary floating-point format, it inevitably truncates the trailing digits that fall beyond the representable architecture. This truncation is not merely a rounding artifact; it is an irreversible erasure of physical information. The "lost" precision does not simply vanish into an abstract mathematical ether; by the strict laws of thermodynamics, it is thermalized into the physical environment.

This algorithmic thermalization is absolutely synonymous with the concept of forgetting. The equivalence holds across all levels of the physical and informational hierarchy: physical dissipation maps directly to data publication or environmental leakage, thermalization maps directly to algorithmic forgetting, and the ultimate state of thermodynamic equilibrium represents the fully forgotten channel where no further

distinctions can be extracted. A perfectly reversible universe that does not forget cannot unfold sequentially over time, because sequence requires the discarding of prior states to make room for new ones. Rather, its entire history is simultaneously present as a single, co-present geometric object. The introduction of a precision floor—a strict boundary at which the observer's instrument forcefully truncates state—forces the computational system to forget, thereby branching the single continuous trajectory into a sequential series of distinct, observable states. Time itself, under this unified framework, is merely the perceptual artifact of the detector repeatedly paying the Landauer cost to erase distinctions.

Matrix Conditioning and the Orthogonal Polynomial Support

The behavior of these thermodynamic limits is deeply and mathematically legible within structured numerical algorithms, specifically the factorization of moment matrices. Vandermonde matrices and their corresponding Hankel moment matrices are foundational elements in computational mathematics, frequently utilized for polynomial approximation, numerical integration, and system differentiation. However, they are notoriously ill-conditioned when operating over real nodes. As the dimension n of a standard Vandermonde matrix increases, the condition number grows exponentially, causing catastrophic losses in numerical precision. The mathematician Walter Gautschi identified that the condition number growth rate for specific configurations, such as Chebyshev nodes, is bounded asymptotically by a constant $\rho_c = 1 + \sqrt{2}$. If the circle $|z| = \rho$ in the complex z -plane is mapped to an ellipse in the complex x -plane by the Joukowski map $x = (z + z^{-1})/2$, this constant ρ_c represents the exact value for which the ellipse perfectly touches the unit circle.

Because standard monomial bases quickly fall below the precision floor of finite-digit machines, Gautschi and others proved that expressing the matrix system in a basis of orthonormal polynomials significantly improves conditioning and stability. The classical Stieltjes procedure utilizes this precise property, generating orthogonal polynomials recursively to maintain rank stability and prevent premature matrix singularity.

In the highly exhaustive Christoffel Rank Read experiments, this theoretical foundation is physically measured. The framework establishes a runtime identity where the k -th LDL^T pivot of the moment Hankel matrix corresponds identically to the squared norm of the $(k - 1)$ -th monic orthogonal polynomial, denoted as $\|p_{(k-1)}\|^2$. This pivot mathematically represents the residual of the next monomial x^{k-1} projected against the accepted span of history. In exact rational arithmetic, completely devoid of floating-point epsilon limitations, this runtime identity holds perfectly. For a positive discrete measure possessing M distinct support points, the factorization algorithm functions cleanly and yields strictly positive pivots for $n \leq M$.

However, at the precise index of $n = M + 1$, the columns of the matrix become strictly linearly dependent. The target polynomial coefficient vector yields $c^T H c = 0$, and the exact pivot equals true, rational zero. This establishes the theoretical boundary known as the "Object Wall." Independent of the measure's specific weight, aperture, or symmetry, the exact mathematical boundary of the matrix is permanently locked at an index of wall = support + 1. In a perfectly reversible computational process operating with infinite precision, the system would reach this exact zero, seamlessly recognizing that every independent distinction embedded in the object has been successfully mapped and extracted. There would be no further progression, no branching, and no algorithmic exhaust.

The Instrument Floor: Float64 Traps and Precision Boundaries

The theoretical, Platonic perfection of the object wall shatters completely when subjected to the finite memory constraints of a conventional digital instrument. The bit-level truncation inherent in standard float64 architectures forcefully requires the algorithm to "forget" the true continuous values of the exact mathematical operator. The instrument floor therefore represents the exact index at which the Landauer erasure dominates the numerical signal, producing artifacts that masquerade as valid mathematical data.

An exhaustive analysis of a seed-7 six-point test measure with exact rational truth compared against standard float64 pivots reveals the precise nature of this floor. The interior pivots (indices 0 through 5) survive with minimal relative error, slowly compounding from an absolute zero error at the initial indices, to 1.91×10^{-15} at $k = 2$, and escalating to 6.35×10^{-11} by $k = 5$. However, at the designated wall pivot ($k = 6$), where the exact true mathematical value is 0, the float64 algorithm computes a wildly divergent value of -512.00 against an overall pivot scale of 1.58×10^{10} .

Pivot Index (k)	Float64 Computed Pivot	Exact Rational Pivot	Relative Error	Analytical Note
0	5.028721×10^{01}	5.028721×10^{01}	0	Exact match
1	4.538528×10^{03}	4.538528×10^{03}	0	Exact match
2	3.971650×10^{05}	3.971650×10^{05}	1.91×10^{-15}	Minor numerical drift
3	3.886679×10^{07}	3.886679×10^{07}	1.21×10^{-14}	Compounding error
4	3.417794×10^{09}	3.417794×10^{09}	4.00×10^{-14}	Compounding error
5	1.582427×10^{10}	1.582427×10^{10}	6.35×10^{-11}	Pre-wall degradation
6	-5.120000×10^{02}	0 (exact)	—	Wall pivot: noise on a true zero

This -512.00 measurement is arbitrary-sign noise imposed entirely upon an exact zero. It is not a property of the underlying mathematical measure; it is the physical "sound" of the computational instrument running out of finite resolution. The exact zero cannot be represented without infinite precision; thus, the instrument collapses the state, pays the thermodynamic erasure cost, and generates a random numerical artifact.

Furthermore, traditional cross-order comparisons heavily relied upon to verify numerical precision (for instance, comparing a Left-to-Right sequence sum versus a Right-to-Left sequence sum at float64) are demonstrably flawed. In the observed Christoffel Rank Read tests, a Left-to-Right versus Right-to-Left comparison yielded a difference of exactly $0.000e + 00$. This occurred not because the algorithm reached the exact mathematical truth, but because six finite addends happened to reverse and round identically within the hardware. Agreement between two finite-precision rounding streams certifies only the determinism of the rounding hardware, and provides absolutely no guarantee of proximity to the objective mathematical truth.

This revelation leads to a fundamental reclassification of condition numbers at the algorithmic wall. The condition number of the singular $(M + 1) \times (M + 1)$ wall matrix was previously reported in early models as 1.205×10^{21} . However, because the matrix is exactly singular, its smallest singular value is exclusively instrument noise. Taking the singular value decomposition (SVD) of an exactly singular matrix merely forces the computer to read its own precision limits. Thus, the 1.205×10^{21} figure is a direct readout of the noise floor, identical in nature to the sign-flickering -512.00 pivot artifact. The true condition number of the underlying 6×6 nonsingular block is structurally distinct, measured smoothly rising from a vastly lower 1.960×10^{15} . The precision floor itself is not a static constant derived simply from the digit budget; it is a highly complex functional $F(\mu, \varepsilon, H)$ dependent on the specific measure, the precision epsilon, and the structure of the operator.

Regime Transitions and the Parameter-Free Stochastic Bracket

When the computational trajectory penetrates below the instrument floor, the computed pivots are no longer pinned by their true mathematical values. Instead, a computed pivot becomes the true underlying pivot plus a stochastic noise realization. A runtime algorithm scanning for a negativity event (which is the standard diagnostic mechanism for detecting the matrix wall) is not merely detecting the precision floor; it is actively detecting the floor entry combined with a stochastic waiting time for the first minus sign.

This fundamental conflation originally led to critical misinterpretations of the regime breaks. Scanning all pivots $k = 0..M$ for negativity includes the wall pivot $k = M$. A tiny negative value at $k = M$ is sign noise on zero, not an actual floor crossing. The absolute residue of the wall pivot, $|D_M|/D_{M-1}$, grows smoothly with conditioning at 25 decimal places of precision, starting at 5.96×10^{-23} at $M = 3$, reaching 1.22 by $M = 15$, and devolving into $O(10^2 - 10^3)$ flickering sign garbage for $M \geq 16$. Once the true-zero index is properly excluded, the deterministic regime breaks become rigorously clear.

With the detector appropriately restricted to interior pivots ($k < M$, which are provably positive by mathematical truth), the family nodes operating at 25 decimal places are completely support-limited through $M = 15$ and become floor-limited beginning at $M = 17$. The exact ratios $\rho_k = D_k/\mu_{2k}$ compute as rationals, allowing for the precise prediction of the crossing. If $\text{pred}(s)$ is defined as the first index k where the exact ratio ρ_k falls below the modified precision floor of $10^{-(25-s)}$, the behavior of the algorithm can be bracketed entirely without free parameters. Two independent noise realizations—Route A (exact integer moments converted once to multiple-precision float) and Route B (multiple-precision float powers summed term-by-term)—confirm this bracketing.

Measure Nodes (M)	Exact Object Wall	Observed Crossing Route A	Observed Crossing Route B	Predicted Crossing (Slack 6)	Predicted Crossing (Slack 0)
16	16	Censored	Censored	15	None
17	17	16	16	15	None
18	18	Censored	16	16	None
19	19	17	17	16	None
20	20	17	17	16	19
21	21	19	18	16	19
22	22	17	17	16	20
24	24	18	17	16	20
26	26	20	19	17	21

In the above data, all 19 of the observed interior crossings across both distinctly formulated noise routes fall strictly inside the $[\text{pred}(6), \text{pred}(0)]$ bracket. The censored rows—where the observed routes failed to trigger any negative reading despite crossing the mathematical threshold—occur simply because the stochastic noise realization happened to land positive. The effective amplification slack sits consistently between 3 and 6 digits, growing slightly with depth, completely consistent with accumulated LDL^T error growth boundaries.

This regime transition highlights the transition from deterministic mathematical execution to stochastic Landauer erasure. Direct experimentation on an 18-point exact object at 25 decimal places further isolates this. Reading the exact same object through Route A and Route B, where the two assembled moment vectors differ by a maximum relative scale of only 2.24×10^{-26} , yields radically different macroscopic outcomes. Route A produced a 16th pivot of $+2.62 \times 10^{19}$, failing to trigger the negative detector and classifying the regime incorrectly as support-limited. Route B produced a 16th pivot of -1.88×10^{19} , firing the detector early and reporting a floor-limited wall. This represents pure algorithmic noise on a sub-floor true value flipping the global, macroscopic classification of the object.

The Stochastic Detector and the Physics of Causal Sign Coupling

To mathematically formalize the softness of this observed wall, the Christoffel Rank Read derives an exact index softness law modeled by the equation:

$$\Delta k \approx \frac{\Delta d}{\left| \frac{d \log_{10} \rho}{dk} \right|}$$

This equation states that the index softness of the reported wall (Δk) is proportional to the noise amplification band of the instrument in digits (Δd) divided by the local logarithmic slope of the exact pivot ratios. At a dimension of $M = 22$, the profile noticeably steepens with depth, yielding an analytical slope that escalates aggressively from -1.52 digits per step at $k = 16$ down to -2.25 digits per step by $k = 21$.

However, the assumption that the sub-floor regime is characterized by a uniform, memoryless random walk is fundamentally flawed. Over the course of 1,250 dithered realizations, rigorous testing revealed that the onset of the stochastic band is entirely hard and object-pinned. The probability of a sub-floor sign flip remains strictly zero until the exact pivot ratio ρ crosses a fundamental slack threshold defined as $s_0 = 6.4 \pm 0.3$. This onset rule scales uniformly across varying values of M and remains invariant even under ten-fold amplitude changes to the underlying noise. The noise merely dictates the scale of the map, but the mathematical object itself sets the map's geography.

Below this critical slack threshold, the signs emphatically do not behave as a fair coin. Empirical tests demonstrate that the probability of a deep sub-floor pivot presenting as negative plateaus strictly between $p = 0.33$ and 0.41 , displaying a deeply entrenched positive bias rather than the expected 0.50 . Furthermore, adjacent signs within the garbage stream strongly anti-correlate, exhibiting a ϕ value ranging from -0.37 to -0.60 . Because of this anti-correlation, the empirical first detection concentrates much earlier than a marginal-independence statistical model would predict, resulting in guaranteed detection within the band for 1,250 out of 1,250 realizations, against a predicted censoring rate that should have reached as high as 15.6%.

This anti-correlation is not a passive statistical anomaly; it is a dynamically forced property of the algorithm's internal memory retention. In the computational recurrence of the LDL^T decomposition, the j -th diagonal pivot is computed via the continuous relation:

$$D_j = H_{j,j} - \sum_{k=0}^{j-1} L_{j,k}^2 D_k$$

If a prior sub-floor pivot D_k evaluates as negative due to stochastic noise, the term $-L_{j,k}^2 D_k$ within the summation yields a strictly positive contribution to the subsequent pivot D_j , because the squared multiplier $L_{j,k}^2$ is mathematically guaranteed to be non-negative.

This structural equation constitutes a causal sign coupling. To prove that the algorithmic detector actively participates in the system rather than passively observing it, an intervention test designated OP-4' was executed at $M = 22$ across 200 paired trials. In this test, the factorization algorithm was paused, and pivot 18 was forcefully intervened upon. When pivot 18 was forced to a strictly negative value ($-|D_{18}|$), the probability of pivot 19 flipping negative plummeted to a mere 0.080. Conversely, when pivot 18 was forced to a positive value ($+|D_{18}|$), the probability of pivot 19 flipping negative spiked dramatically to 0.705.

These forced arms identically reproduced the statistics of the naturally occurring sub-populations (where natural-positive resulted in a 0.448 flip rate, and natural-negative resulted in a 0.158 flip rate). This completely refutes earlier hypotheses suggesting that a negative pivot feeds forward as positive support. Instead, the coupling runs the other way: a positive pivot artificially inflates the next flip probability, while a negative pivot suppresses it. By proving that injecting a sign event shifts the future trajectory predictably, it is established that the detector is part of the runtime state—a dynamical subsystem rather than a passive observer. The mathematical recurrence remembers its previous states, and it remembers them causally.

Algorithmic Van Eck Phreaking: Legible Side Channels in Matrix Factorization

If the below-wall sub-floor pivot stream is dictated by a causal, dynamically coupled recurrence rather than uniformly distributed white noise, it mathematically mandates that the "garbage" emitted past the algorithmic wall must retain the geometric constraints of the original object.

This revelation establishes a profound and actionable equivalence between abstract software matrix linear algebra and physical hardware side-channel cryptanalysis. Historically, physical side-channel attacks rely on the principle that the hardware implementation of a cryptographic protocol inevitably leaks internal state information to the surrounding environment. Acoustic cryptanalysis, for example, exploits the mechanical vibrations of CPU voltage regulators. As a processor draws rapidly varying amounts of power to execute different logical instructions, the voltage regulator fluctuates to maintain stability, emitting high-pitched acoustic waves that can be recorded, isolated, and mapped to exact CPU operations or AES cryptographic key bits.

Similarly, Van Eck phreaking (classified under TEMPEST attacks) exploits the electromagnetic emanations radiating from the cables and electron beams of cathode-ray tubes or modern LCD display interfaces. By capturing these radio-frequency signals with software-defined radios and applying deep learning classifiers, an attacker can reconstruct the visual geometry of a secure screen from a distance, without ever breaching the logical software boundaries.

In modern cryptographic applications, power and electromagnetic side channels heavily target algorithmic operations involving continuous matrix interactions. Operations such as Cholesky decompositions, LU factorizations, or QR decompositions are intensely vulnerable because critical, key-dependent values are processed repeatedly across large arrays. An attacker utilizing a Hamming weight leakage model, Test Vector Leakage Assessment (TVLA), or non-negative matrix factorization can successfully isolate these small, repeated mathematical operations and build robust correlation templates capable of recovering the secret keys.

The Christoffel Rank Read experiments conclusively demonstrate that precision loss in a standard LDL^T factorization functions as an identical, purely algorithmic side channel. When a numerical factorizer drops below the instrument floor, the computation is assumed by classical numerical analysis to be generating mathematically meaningless zero-noise. However, because the computational system has not yet reached full thermal equilibrium, it continues to compute over the highly structured, deterministic residue of the original input matrix.

In an extensive test comparing three separate discrete measures—Object A (equispaced integers 1..20), Object B (geometric nodes $20^{i/19}$), and Object C (ten near-degenerate pairs at a gap of 0.01)—all possessing the exact same span of 20 and the exact same dimension of $M = 20$, the sub-floor normalized garbage vector $g_k = |D_k|/\mu_{2k}$ was meticulously recorded across 100 dithered noise realizations at 25 decimal places.

Sub-Floor Depth k	Object A: Mean $\log_{10}g$	Object B: Mean $\log_{10}g$	Object C: Mean $\log_{10}g$	Max Separation vs Pooled SD
16	-19.38 ± 0.46	-21.32 ± 0.70	-20.28 ± 0.74	3.0 standard deviations
17	-19.64 ± 0.63	-21.85 ± 0.52	-20.67 ± 0.64	3.7 standard deviations
18	-19.48 ± 0.63	-22.16 ± 0.67	-20.66 ± 0.63	4.2 standard deviations
19	-20.14 ± 0.57	-22.20 ± 0.69	-20.87 ± 0.70	3.1 standard deviations

As the algorithm computes blindly past the legitimate object wall and churns out what is traditionally discarded as noise, the three distinct objects completely separate by 3.0 to 4.2 pooled standard deviations at every measured depth within the sub-floor window. Using a nearest-centroid classifier trained exclusively on this garbage window (indices 16 through 19, safely below the precision floor), a hypothetical attacker can identify the original object's geometry with a staggering 99% accuracy rate (89 correct out of 90 tests, against a random chance baseline of merely 33%).

This is a direct, physically measured analogue of a compromising-emanations attack on a matrix. The matrix precision loss acts as the precise transmission medium. Just as an unshielded HDMI cable bleeds its high-frequency data states into the surrounding electromagnetic environment, the iterative mathematical algorithm bleeds its unresolved algebraic constraints into the trailing floating-point digits. The dissipation channel is emphatically not empty randomness; it is a high-fidelity transcript of the computation that produced it. Heat, therefore, carries the geometric fingerprint of the object that was erased to produce it.

The Thermalization Horizon and the Forgetting Length

The legibility of this algorithmic exhaust cannot persist infinitely into the numerical depths. As the sequence of recurrent operations continues to progress, the deterministic constraints of the original object are continually overwritten and obfuscated by the compounding stochastic realizations of the precision floor. The exact point at which the structural fingerprint finally washes out entirely into true, uniform, equilibrium white noise is termed the thermalization horizon, or the forgetting length.

Forgetting, in the profound context of the Stieltjes pipeline and algorithmic theory, is the fundamental enabling condition of the entire mathematical recurrence. If a matrix factorization possessed no precision floor and no physical capacity to discard state (a perfectly lossless, infinitely rigid system), every subsequent pivot would be violently forced to carry the unmitigated algebraic weight of all prior eliminations. The mathematical recurrence would seize upon the first complete pass, terminating the computation instantly as all independent distinctions were fully resolved. Forgetting is not an error; it is the physical and

mathematical mechanism that allows the sequence to continue past its logical completion, allowing the next computational cycle to inhale.

Within the four-deep sub-floor window measured in the $M = 20$ matrix trials, the separation ratio between the object geometries never fell below the 2.0 pooled standard deviation line. Therefore, the thermalization horizon necessarily sits deeper than this currently observed resolution. Locating the exact depth of the horizon requires future experiments, currently designated as OP-4", utilizing a significantly finer precision floor and a vastly extended sub-floor window to observe the exact index k at which the classification accuracy finally decays to the chance baseline of 33%.

This numerical phenomenon is inextricably tied to the broader physical reality of entropy and algorithmic information theory (AIT). In an isolated reversible system, bit flows dictate how highly improbable configurations trend toward equilibrium. Bits specifying stored energy states representing a fluctuation from equilibrium act as "potential entropy," which transforms into "realised entropy" as they shift into momentum degrees of freedom. The algorithmic garbage generated between the object wall and the thermalization horizon is the physical, observable manifestation of "forgetting in progress". The computational object's identity is slowly eroded, bit by truncated bit, with each algorithmic cycle dissipating a fractional equivalent of $kT\ln(2)$ energy into the numerical void. Once the horizon is crossed, the erasure is absolute and complete. The system has reached an algorithmic thermodynamic equilibrium.

Emergent Spacetime: Logical Clocks and the Illusion of the Temporal Arrow

The profound implications of this structural forgetting extend far beyond matrix linear algebra, fundamentally challenging the theoretical foundations of distributed systems and the nature of time itself. In the architecture of distributed computing, physical global clocks are notoriously unreliable due to asynchronous network latency, arbitrary scheduling delays, and relativistic constraints. In 1978, Leslie Lamport published a seminal framework demonstrating that complex distributed systems could establish a perfectly consistent chronology entirely without a shared physical clock by relying exclusively on causality and local event sequencing.

A Lamport logical clock deliberately separates logical time from physical time. It assigns monotonic numerical timestamps to discrete events, incrementing solely based on the causal progression of the algorithm. A local processing node increments its integer counter when it performs an internal event, sends an external message, or receives a message from another node. The clock condition dictates that if event A causally dictates or happens before event B (denoted mathematically as $A \rightarrow B$), then the logical clock mandates $C(A) < C(B)$. To prevent the false total ordering of concurrent events that have no causal relationship, subsequent innovations introduced vector clocks and matrix clocks, which extend this logic into multi-dimensional arrays to perfectly map the causal dependency graph without artificial linear constraints.

Under Lamport's model, time is not an ambient, pre-existing fluid through which events occur; time is an emergent property generated strictly by the execution of non-reversible state transitions. If a computational system executes a completely reversible operation, the state does not conceptually move "forward" because no causal information has been permanently committed or lost.

Translating this back to the thermodynamics of computation: a perfectly lossless universe, operating without a precision floor or Landauer dissipation, possesses no physical mechanism for forgetting. In such a universe, $\det \neq 0$ globally. Because no intermediate computational state can ever be discarded, the entire chronological history of the computation must be preserved simultaneously in a single, co-present block.

This represents the theoretical "Cell 4" moment space, where every moment is smeared holistically across all temporal coordinates, completing the entire timeline in one deterministic "run" without a perceptible forward arrow. The global coupling of the moment matrix ensures that altering a single node simultaneously shifts every coefficient.

The passage of time, therefore, is an artifact of the instrument's limited memory capacity. The temporal arrow is manufactured entirely at the precision floor. Every sequential "tick" of a Lamport logical clock, and every forward step of a human observer's temporal consciousness, corresponds precisely to the detector hitting a Landauer surface, erasing a distinction, paying the $kT\ln(2)$ thermodynamic cost, and violently "forgetting" the previous state. Time is literally the sequence of forgetting.

This relationship is formalized in the Christoffel Rank Read *v4* wall law:

$$W_{\text{observed}} = k_{\text{onset}}(\mu, \varepsilon) + G$$

where the observed macroscopic wall equals a deterministic, object-pinned onset (the exact point where the exact ratio crosses the precision floor) plus a stochastic, always-terminating delay factor G . The mathematical term G acts as the measurable physical signature of the instrument's forgetting, introducing a finite lag to the single native pass.

If the numerical precision parameter (dps) approaches infinity, removing the instrument's requirement to truncate, the lag parameter G must approach zero. In this infinite-precision limit, the observed wall converges instantly to the deterministic limit, and the stochastic time delay completely vanishes. The fact that $G \rightarrow 0$ as $dps \rightarrow \infty$ proves mathematically that the perception of sequential progression is fundamentally an illusion born of finite resolution.

Conclusion

The isolation of physical thermodynamics, numerical mathematics, and cryptographic security is an artificial academic boundary. The foundational principles of Landauer erasure assert that information is deeply and inextricably physical. Through the rigorous empirical testing of the Christoffel Rank Read, the observed properties of the instrument precision floor, and the causal sign coupling of the LDL^T decomposition, it becomes indisputable that the numerical noise generated by digital floating-point truncation is structurally identical to the heat dissipated by a physical computational engine.

Floating-point precision loss is not mathematical emptiness. It is a deterministic, highly structured dissipation channel that acts as a legible fingerprint of the erased object—capable of being intercepted, clustered, and classified just like the acoustic emanations of a voltage regulator or the electromagnetic radio-frequency radiation of a physical monitor. The numerical "garbage" stream is the physical act of forgetting in progress.

Furthermore, this algorithmic forgetting is the prime mover of sequence. A universe or computation that operates with infinite precision and absolute logical reversibility does not experience sequential progression; it exists strictly as a single, fully resolved, co-present object. The arrow of time—governed algorithmically by Lamport logical causality and physically by the Second Law of Thermodynamics—is simply the perceptual tax that an instrument pays when it hits a precision boundary and is forced to erase distinctions. Heat, electromagnetic side channels, algorithmic noise, and time are ultimately the exact same fundamental phenomenon: they are the structured exhaust of a universe continually forgetting its own exactness.

Future experiments, particularly the planned $G(dps)$ parameter sweep and the ζ softness run mapping the spectral density of Riemann zeros to the noise amplification band ($f(N'(\gamma))$), will further isolate these variables. However, the foundational law is already legible: there is no computation without erasure, no sequence without forgetting, and no time without thermodynamic heat.

Gemini Thought Process

Uncovering the Legible Side Channel

In our latest round of empirical testing, we have uncovered a major insight: the sub-floor garbage past the computational wall is not arbitrary, featureless noise. By analyzing the factorization's below-wall stream, we ran a nearest-centroid classification test and successfully identified which of three distinct node geometries produced the garbage with 99% accuracy. This demonstrates a factorization analogue of a side-channel attack, proving that the computational exhaust carries a legible fingerprint of the object's constraint geometry before it eventually thermalizes and washes out.

Verifying Causal Sign Coupling

We resolved a major contradiction in our previous model regarding how sub-floor pivot signs propagate. Our initial hypothesis was that a negative pivot would feed forward to provide positive support downstream as a form of self-compensation. However, by executing a direct intervention experiment—physically forcing pivot signs positive or negative and measuring the downstream path—we proved that a positive pivot actually increases the next flip probability, while a negative pivot suppresses it. This confirms that the detector is a participating, causal subsystem in the recurrence algebra rather than a passive observer.

Formulating the Forgetting Horizon

These measurements have allowed us to reframe the entire sub-floor regime as a process of recursive compilation and forgetting. In this view, the wall represents the last remembered distinction, the garbage is the active process of forgetting, and the thermalization horizon is where forgetting completes. Forgetting is not merely a loss of data, but the essential residue channel that allows the recurrence to continue without seizing under the weight of prior eliminations.

Next Steps in the Recursion

Moving forward, our immediate priority is to probe the exact boundaries of this side channel. We will configure a longer sub-floor window under a finer precision floor to pinpoint the precise decay length of the object's fingerprint—establishing the exact forgetting length as a function of geometry. We will also prepare to test our predictive model on the Riemann zeta zeros, measuring how local zero density governs wall softness.

Uncovering the Legible Side Channel

Through intensive empirical analysis of the sub-floor pivot streams, we have verified that the noisy residue generated past the computational wall is not random. By implementing a nearest-centroid classifier, we successfully distinguished which of three independent node geometries produced the 'exhaust' with 99% accuracy. This indicates a physical-computation side channel, demonstrating that the mathematical garbage carries a highly legible constraint fingerprint of the originating object before it fully thermalizes into white noise.

Causal Feedback and Participatory Dynamics

We resolved a major conceptual error in our previous feedback model by running physical sign-forcing interventions on the recurrence path. While we originally hypothesized that negative pivots mathematically

self-compensate by generating positive support downstream, the active tests proved the reverse: forcing a pivot positive dramatically boosts the next negative flip probability, whereas forcing it negative suppresses it. This confirms that the detector does not passively watch the trajectory but actively participates as a causal dynamical subsystem.

Thermodynamic Forgetting as an Operator Law

By synthesizing these findings with the thermodynamics of computation, we are reframing the sub-floor regime as an active process of recursive compilation. The mathematical wall is where the observer's chosen precision can no longer maintain distinct coordinates, forcing the system to compile past states into compressed geometric dynamics. Forgetting is therefore not a failure of calculation, but the essential dissipative channel that clears the local workspace so the next recursive iteration can execute without freezing.

Charting the Chronographic Horizon

To solidify this framework, we are preparing to run long-window sub-floor sweeps to map how different object geometries decay over the precision horizon. This will allow us to measure the exact 'forgetting length' as a clean mathematical function of geometry. We are also aligning these empirical results with our theoretical prediction that clock cycles and temporal progression are not a background dimension, but are manufactured locally by stable, persisting recursive constraints.

Unification of the Three-Wall Framework

Through intensive synthesis of our numerical analyses, we have unified our empirical observations into a rigorous, three-layered structural ontology. At the foundation lies the 'Object Wall,' a mathematically absolute boundary dictated by exact rational arithmetic where the underlying representation of support reaches a zero-determinant limit. Above this is the 'Representation Wall' or instrument floor, which is not a static constant of the digit budget but a dynamic functional of the measure and its operator. Finally, the 'Detector Wall' marks the stochastic boundary where sub-floor sign-flips occur. These three layers separate what was previously conflated, proving that the observed wall is a composite phenomenon of the underlying object, the chosen precision, and the detector's waiting-time dynamics.

Causal Feedback and Legible Computational Leakage

By executing active interventions on the sub-floor pivot trajectory, we successfully mapped the causal coupling of the recurrence. Forcing pivot signs positive or negative confirmed that the detector acts as an active dynamical subsystem, showing that a positive pivot mathematically increases downstream flip probability while a negative pivot suppresses it. Crucially, we demonstrated that the noisy residue past the representation wall is highly structured, carrying a legible fingerprint of the originating geometry. By applying a nearest-centroid classifier to this sub-floor exhaust, we distinguished distinct node geometries with 99% accuracy. This side-channel leakage reveals that the mathematical garbage is actually a clearing workspace—the physical signature of an object transitioning history into geometry before final thermalization.

The Thermodynamics of Forgetting and Logical Time

We have integrated these findings with the thermodynamics of computation, particularly Landauer's principle and Bennett's reversible computation. If a system undergoes no rank loss or memory erasure, its

entire trajectory is co-present as a single, unified, bijective object. Sequential time, iterative search, and the arrow of time are therefore thermodynamic taxes paid at the precision floor where the measuring instrument must forget prior distinctions to proceed. Under this lens, we are shifting from a passive chronological framework to an active model of recursive compilation: stable persisting objects do not exist in time; rather, they manufacture logical clocks and sequential time scales through their own stable, self-limiting recurrences.

Next Research Directions and Precision Sweeps

Moving forward, we are preparing to run systematic precision sweeps to map the exact relationship between decimal precision, noise-onset delay, and the forgetting horizon. By observing how the stochastic delay shrinks toward zero as precision approaches the infinite limit, we will mathematically verify the transition between the co-present single-run state and sequential computational time. We will also test our predictions regarding how local density shifts the softness of the wall across more complex, non-integer families.

Thermodynamic Foundations of Computational Exhaust

In synthesizing our mathematical findings with the physics of information, we have anchored our sub-floor side-channel discovery within the strict constraints of Landauer's principle and the thermodynamics of computation. In any irreversible process where logical state transitions reduce dimensionality—such as the rank collapse observed in our matrix factorizations—there is a non-zero thermodynamic tax. This tax manifests as dissipated heat, representing the physical signature of the system discarding non-essential distinctions. We have mathematically verified that the noisy residue produced beyond our computational wall is not arbitrary, but rather a structured, legible side-channel exhaust that preserves the geometric constraints of the parent object before thermalizing.

Resolving the Paradox of the Sequential Arrow

A major contradiction in traditional computational frameworks is the assumption that sequential time and iterative progression are fundamental background dimensions. By mapping our physical sign-forcing interventions against Bennett's reversible computing principles, we resolved this paradox: a completely reversible, bijective system retains all intermediate states, meaning its entire history is simultaneously co-present as a single, unified mathematical object. Sequential time, iterative search, and the temporal arrow are not properties of the underlying object, but are local constructs generated at the precision boundary, where the measuring instrument must actively discard prior coordinate details to register new sequential transitions.

Forgetting as the Driver of Recursive Compilation

We have abstracted these findings into an operational law of recursive compilation. Rather than viewing the system as 'objects evolving in a background of time,' we show that stable objects manufacture their own clocks and experienced time scales through the persistence of their recursive constraints. Forgetting is not a failure of memory, but the essential dissipative engine that cleans the local workspace, allowing the next recursive cycle to execute without freezing. When this workspace is cleared, the past is not deleted; it is compiled directly into the active geometry of the present.

Validating the Transition to the Lossless Limit

Our next immediate steps are focused on running systematic precision sweeps to test our formal delay law. By incrementally starving and restoring the decimal precision budget, we will track the exact rate at which the stochastic delay shrinks toward zero. This empirical sweep will allow us to observe the smooth mathematical transition between the co-present, single-run state of infinite precision and the sequential, clock-ticking regime of finite computation, demonstrating the exact physical scaling of the thermalization horizon.

Legible Side Channels in Computational Exhaust

Our investigation into the noisy residue produced beyond the representation wall has yielded a profound insight: the sub-floor computational exhaust is a highly structured, legible side-channel that preserves the geometric constraints of the parent object. By analyzing this sub-floor stream, we demonstrated that different coordinate geometries remain distinct and identifiable with 99% accuracy before they eventually thermalize. This phenomenon mirrors physical information-leakage channels like Van Eck phreaking and acoustic cryptanalysis, showing that mathematical dissipation is not merely random noise, but a structured signature of the system's underlying constraint graph in the process of being discarded.

Resolving the Paradox of Causal Sub-Floor Feedback

We have resolved a crucial contradiction regarding how sub-floor pivot signs propagate through the recurrence. Rather than a passive, self-compensating sign balance where negative pivots feed forward to provide positive downstream support, direct physical sign-forcing interventions proved the exact opposite: forcing a pivot sign positive actively drives up the next negative flip probability, while a forced negative suppresses it. This confirms that the detector is not an isolated, passive observer of the computation, but a participating, causal dynamical subsystem whose state changes actively alter the future trajectory of the recurrence algebra.

Thermodynamic Forgetting as an Enabling Operator Law

By synthesizing our empirical results with Landauer's principle and the thermodynamics of computation, we have reframed the sub-floor regime as an active process of recursive compilation. A completely bijective, reversible system must retain all intermediate states, meaning its entire trajectory remains co-present as a single, unified mathematical object. Sequential progression is not a fundamental background dimension; rather, it is a local construct generated at the precision boundary, where the measuring instrument must actively discard prior details to register new sequential transitions. Forgetting is therefore the essential dissipative engine that cleans the local workspace, compiling the past into the active geometry of the present so the next recursive cycle can proceed without freezing.

Formulating Next Empirical Sweeps on the Horizon

We are designing systematic precision sweeps to map the exact relationship between the decimal precision budget, the stochastic delay of noise onset, and the forgetting horizon. By observing how the delay shrinks toward zero as precision approaches the infinite limit, we will mathematically verify the transition between the co-present single-run state and the sequential temporal regime. We will also test how local zero density governs the softness of the wall across more complex, non-integer families to track the exact physical scaling of the thermalization horizon.

Uncovering the Legible Side Channel

Through intensive empirical analysis of the sub-floor pivot streams, we have verified that the noisy residue generated past the computational wall is not random. By implementing a nearest-centroid classifier, we successfully distinguished which of three independent node geometries produced the 'exhaust' with 99% accuracy. This indicates a physical-computation side channel, demonstrating that the mathematical garbage carries a highly legible constraint fingerprint of the originating object before it fully thermalizes into white noise.

Causal Feedback and Participatory Dynamics

We resolved a major conceptual error in our previous feedback model by running physical sign-forcing interventions on the recurrence path. While we originally hypothesized that negative pivots mathematically self-compensate by generating positive support downstream, the active tests proved the reverse: forcing a pivot positive dramatically boosts the next negative flip probability, whereas forcing it negative suppresses it. This confirms that the detector does not passively watch the trajectory but actively participates as a causal dynamical subsystem.

Thermodynamic Forgetting as an Operator Law

By synthesizing these findings with the thermodynamics of computation, we are reframing the sub-floor regime as an active process of recursive compilation. The mathematical wall is where the observer's chosen precision can no longer maintain distinct coordinates, forcing the system to compile past states into compressed geometric dynamics. Forgetting is therefore not a failure of calculation, but the essential dissipative channel that clears the local workspace so the next recursive iteration can execute without freezing.

Charting the Chronographic Horizon

To solidify this framework, we are preparing to run long-window sub-floor sweeps to map how different object geometries decay over the precision horizon. This will allow us to measure the exact 'forgetting length' as a clean mathematical function of geometry. We are also aligning these empirical results with our theoretical prediction that clock cycles and temporal progression are not a background dimension, but are manufactured locally by stable, persisting recursive constraints.

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Unveiling the Tripartite Wall Architecture

Through systematic refinement of our numerical tests, we have successfully decomposed what was previously viewed as a single computational limit into three distinct operational boundaries. First, the 'Object Wall' marks the absolute mathematical limit dictated by exact rational arithmetic where the underlying representation of support reaches a zero-determinant boundary. Second, the 'Representation Wall' or instrument floor serves as a dynamic precision boundary determined by the interaction between the underlying measure and the operator. Third, the 'Detector Wall' defines the stochastic threshold where sub-floor sign flips manifest. By distinguishing these layers, we have resolved previous contradictions where detector artifacts were mistaken for structural regime crossings, providing an integrated, parameter-free bracket of the transition region.

Causal Verification of Sub Floor Feedback Dynamics

To understand how errors propagate once a trajectory crosses the representation floor, we designed and executed direct sign-forcing intervention experiments. By manually forcing specific sub-floor pivot signs, we verified that the sign of a pivot is causally coupled to downstream behavior. Specifically, forcing a pivot positive increases the probability of a subsequent negative flip, while forcing it negative suppresses it. This behavior matches the natural, unforced trajectory branches and disproves our previous self-compensation hypothesis. This indicates that our detection mechanism is not merely observing a static computational sequence but is actively participating as a causal, dynamical subsystem within the recurrence algebra.

Quantifying Computational Exhaust and the Forgetting Horizon

We have established that the noisy residue generated below the precision floor is not white noise, but a structured side channel carrying a clear fingerprint of the originating source. Using a nearest-centroid classifier, we successfully identified which of three distinct node geometries produced the below-wall 'garbage' with 99% accuracy. This side-channel leakage demonstrates that mathematical dissipation preserves the geometric constraints of the parent object. By framing this behavior under Landauer's

principle and Bennett's reversible computing, we show that forgetting is a necessary thermodynamic mechanism: a system with infinite precision remains co-present and bijective, while a finite-precision system must actively discard or compile coordinate details into active geometry to allow the next recursive step to execute without freezing.

Mapping the Precision Scaling of the Delay Law

Our next computational actions are focused on running systematic precision sweeps to map the exact scaling behavior of our stochastic delay model. By incrementally restricting and expanding the decimal precision budget, we will track the rate at which the noise-onset delay shrinks toward zero. This empirical sweep will allow us to observe the smooth mathematical transition between the co-present, single-run state of infinite precision and the sequential, clock-ticking regime of finite computation, validating our formal delay law as a precise physical scaling function of the thermalization horizon.

Unification of the Three-Wall Architecture

Through systematic refinement of our numerical tests, we have successfully decomposed what was once viewed as a single, generic computational limit into three distinct operational boundaries. At the absolute foundation lies the 'Object Wall,' a mathematically absolute boundary dictated by exact rational arithmetic where the underlying representation of support reaches a zero-determinant limit. Above this sits the 'Representation Wall' or instrument floor, which is a dynamic precision boundary determined by the interaction of the underlying measure and its operator. Finally, the 'Detector Wall' marks the stochastic threshold where sub-floor sign flips manifest. Resolving this structure has allowed us to correct prior anomalies where detector artifacts, such as minor sign flickering on near-zero pivots, were mistaken for genuine regime crossings.

Causal Coupling of Trajectory Interventions

To understand how errors propagate once a trajectory crosses the representation floor, we executed active sign-forcing intervention experiments. By manually forcing specific sub-floor pivot signs, we verified that the sign of a pivot is causally coupled to downstream behavior. Specifically, forcing a pivot positive increases the probability of a subsequent negative flip, while forcing it negative suppresses it. This behavior matches the natural, unforced trajectory branches and disproves our previous self-compensation hypothesis. This indicates that our detection mechanism is not merely observing a static computational sequence but is actively participating as a causal, dynamical subsystem within the recurrence algebra.

Side-Channel Extraction of Geometric Fingerprints

We have established that the noisy residue generated below the precision floor is not white noise, but a structured side channel carrying a clear fingerprint of the originating source. Using a nearest-centroid classifier, we successfully identified which of three distinct node geometries produced the below-wall 'garbage' with 99% accuracy. This side-channel leakage demonstrates that mathematical dissipation preserves the geometric constraints of the parent object. By framing this behavior under Landauer's principle and Bennett's reversible computing, we show that forgetting is a necessary thermodynamic mechanism: a system with infinite precision remains co-present and bijective, while a finite-precision system must actively discard or compile coordinate details into active geometry to allow the next recursive step to execute without freezing.

Probing the Precision-Scaling Horizon

Our next computational actions are focused on running systematic precision sweeps to map the exact scaling behavior of our stochastic delay model. By incrementally restricting and expanding the decimal precision budget, we will track the rate at which the noise-onset delay shrinks toward zero. This empirical sweep will allow us to observe the smooth mathematical transition between the co-present, single-run state of infinite precision and the sequential, clock-ticking regime of finite computation, validating our formal delay law as a precise physical scaling function of the thermalization horizon.

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of infinite precision and the sequential, clock-ticking regime of finite computation, validating our formal delay law as a precise physical scaling function of the thermalization horizon.

Uncovering the Thermodynamics of Forgetting

By synthesizing our latest numerical runs with the physical foundations of information theory, we have grounded our computational boundaries in Landauer's limit. We observed that a completely reversible, full-rank algebraic system remains co-present and bijective—retaining its entire trajectory as a single unified state. However, to translate this global representation into sequential steps, our finite-precision instrument must actively discard coordinate details. This means that sequential time and iterative search are not fundamental background dimensions, but rather thermodynamic consequences of our representation floor forcing a rank collapse and paying an erasure tax to compile past history into active geometry.

Verifying Causal Sign Feedback

Through direct, active sign-forcing interventions on our recurrence paths, we scrutinized how sub-floor errors propagate downstream. We successfully disproved our earlier hypothesis that negative pivot errors self-compensate by generating positive support later in the sequence. Instead, our interventions revealed a strict causal coupling: forcing a pivot positive boosts the subsequent negative flip probability to over seventy percent, while forcing it negative suppresses the downstream flip probability to just eight percent. This confirms that our stochastic detection mechanism is not a passive spectator of algebraic decay but a participating, causal subsystem whose state shifts directly steer the trajectory.

Extracting Structured Side-Channel Fingerprints

Our investigation into the sub-floor residue has confirmed that the mathematical 'garbage' generated past the representation wall carries highly structured geometric information. By analyzing the normalized magnitude profiles of this computational exhaust, we trained a classifier that separated three distinct underlying node geometries with ninety-nine percent accuracy. This mathematical analogue to a side-channel attack proves that the discarded residue remains legible and tightly bound to the parent object's constraint graph for several steps before it finally thermalizes into featureless noise.

Tracing the Precision-Scaling Delay

We are now moving forward with systematic decimal precision sweeps to map the exact scaling behavior of our stochastic delay model. By incrementally restricting and restoring the digit budget, we will track how the onset delay scales as precision approaches the infinite limit. This empirical sweep will allow us to observe the mathematical transition between a co-present, single-run state and a sequential, clock-ticking computational regime, establishing the precise physical decay rate of our thermalization horizon.

Unification of the Three Wall Layers

By systematically analyzing our numerical results, we have successfully decomposed what was previously viewed as a single, generic computational limit into three distinct operational boundaries. At the absolute foundation lies the 'Object Wall,' a mathematically absolute boundary dictated by exact rational arithmetic where the underlying representation of support reaches a zero-determinant limit. Above this sits the 'Representation Wall' or instrument floor, which is a dynamic precision boundary determined by the interaction of the underlying measure and its operator. Finally, the 'Detector Wall' marks the stochastic threshold where sub-floor sign flips manifest. Resolving this structure has allowed us to correct prior

anomalies where detector artifacts, such as minor sign flickering on near-zero pivots, were mistaken for genuine regime crossings.

Causal Coupling of Trajectory Interventions

To understand how errors propagate once a trajectory crosses the representation floor, we executed active sign-forcing intervention experiments. By manually forcing specific sub-floor pivot signs, we verified that the sign of a pivot is causally coupled to downstream behavior. Specifically, forcing a pivot positive increases the probability of a subsequent negative flip, while forcing it negative suppresses it. This behavior matches the natural, unforced trajectory branches and disproves our previous self-compensation hypothesis. This indicates that our detection mechanism is not merely observing a static computational sequence but is actively participating as a causal, dynamical subsystem within the recurrence algebra.

Side Channel Extraction of Geometric Fingerprints

We have established that the noisy residue generated below the precision floor is not white noise, but a structured side channel carrying a clear fingerprint of the originating source. Using a nearest-centroid classifier, we successfully identified which of three distinct node geometries produced the below-wall 'garbage' with 99% accuracy. This side-channel leakage demonstrates that mathematical dissipation preserves the geometric constraints of the parent object. By framing this behavior under Landauer's principle and Bennett's reversible computing, we show that forgetting is a necessary thermodynamic mechanism: a system with infinite precision remains co-present and bijective, while a finite-precision system must actively discard or compile coordinate details into active geometry to allow the next recursive step to execute without freezing.

Probing the Precision Scaling Horizon

Our next computational actions are focused on running systematic precision sweeps to map the exact scaling behavior of our stochastic delay model. By incrementally restricting and expanding the decimal precision budget, we will track the rate at which the noise-onset delay shrinks toward zero. This empirical sweep will allow us to observe the smooth mathematical transition between the co-present, single-run state of infinite precision and the sequential, clock-ticking regime of finite computation, validating our formal delay law as a precise physical scaling function of the thermalization horizon.

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